Jackfruit/Level 3 Problem:

Figure shows a Figee floating crane with a lemniscate boom configuration, and Fig. P11.13b shows a schematic diagram of the crane with dimensions given in the legend. The lifting capacity is 16 T (where 1 T = 1 metric ton = 1 000 kg) including the grab, which is about 10 T. The maximum outreach is 30 m, which corresponds to the position θ2 = 49◦. The minimum outreach is 10.5 m at θ2 = 132◦. For the maximum outreach posture and a grab load of 10 T (under standard gravity), find the bearing reactions at A, B, O2, and O4, as well as the torque, T12, at O2. Notice that the photograph shows a counterweight on link 2; neglect this weight and also the weights of the members.

(a)

(b)

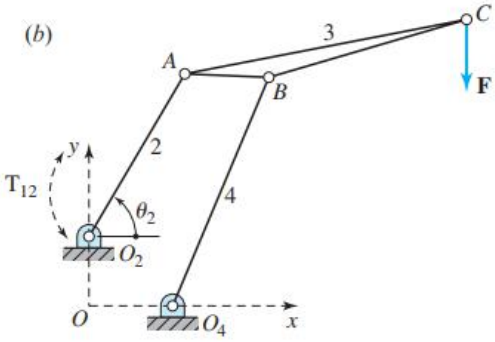
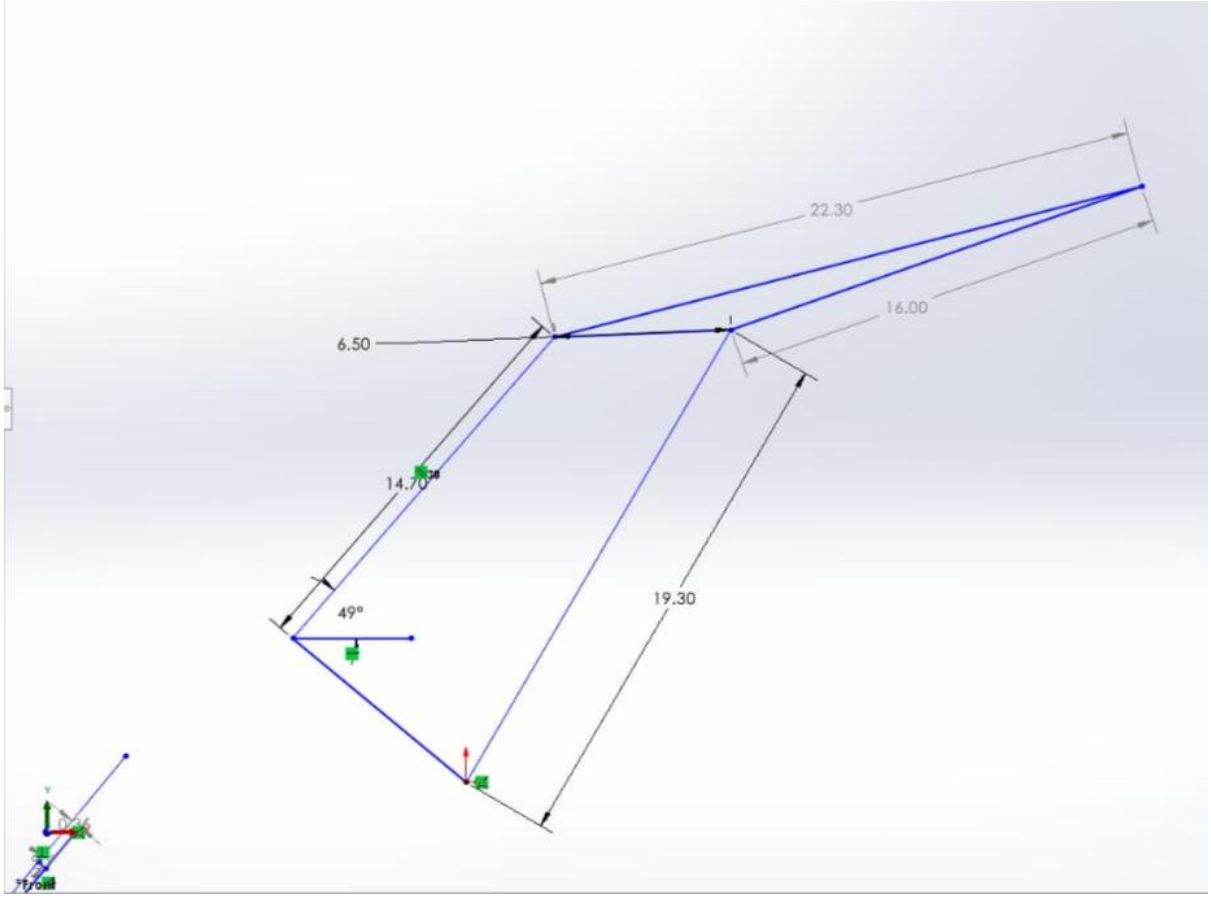
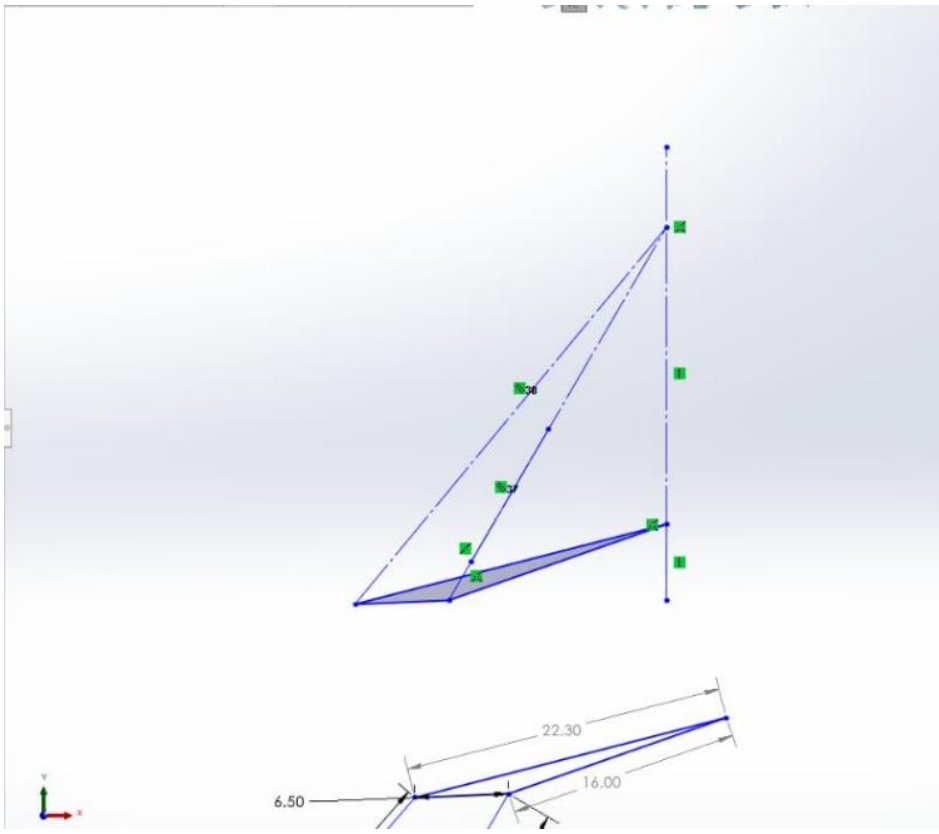


Figure P11.13 (*a*) Photograph; (*b*) $R_{AO_2} = 14.7$ m, $R_{BA} = 6.5$ m, $R_{BO_4} = 19.3$ m, $R_{CA} = 22.3$ m, $R_{CB} = 16$ m, and $\mathbf{R}_{O_2O_4} = -6.4\hat{\mathbf{i}} + 5.3\hat{\mathbf{j}}$ m. (Courtesy of B.V. Machinefabriek Figee, Haarlem, Holland).

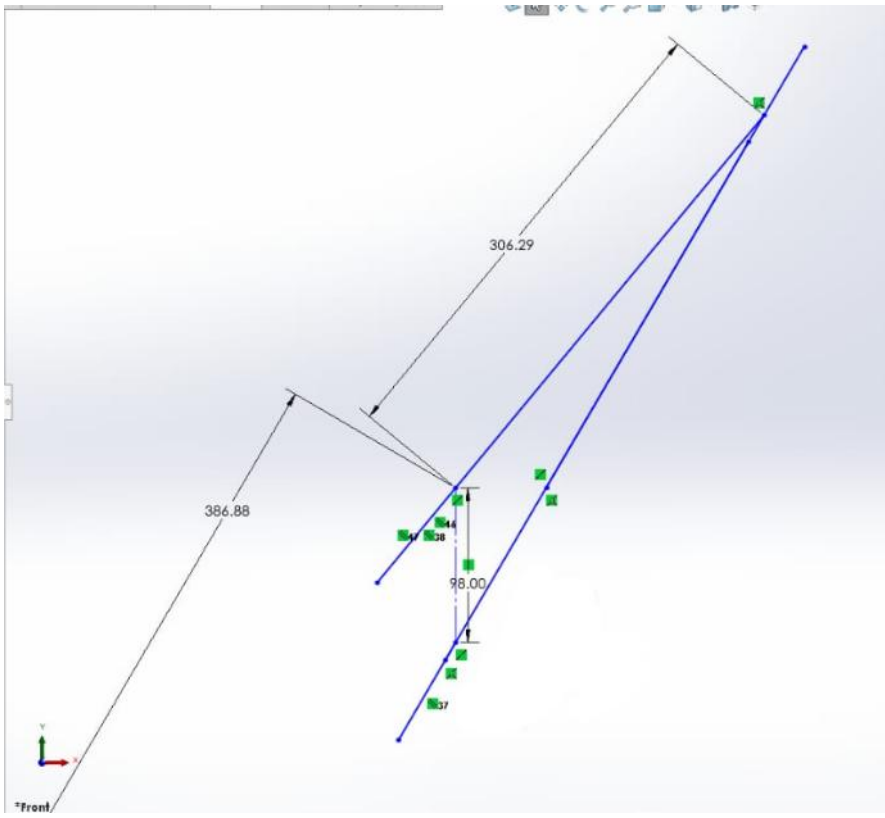
Configuration Diagram: (Units in metres or Newtons)





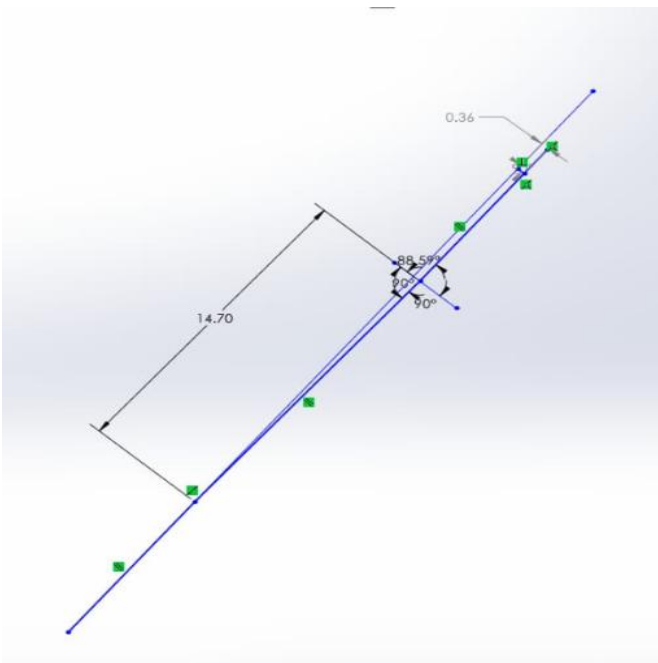
Line of Action :

Static Force Analysis:



Force = 306.29N

Torque Analysis:



Perpendicular distance = 0.36m PTO……

Torque Calculations :

$T = F * h$

$T = 306.29 * 0.36$

Torque = 110.2644 N-m